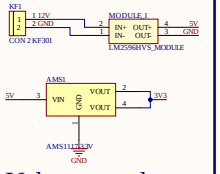
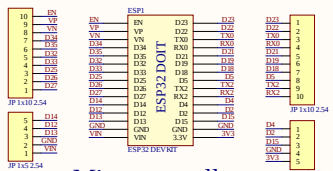
Controller Node



Voltage regulator

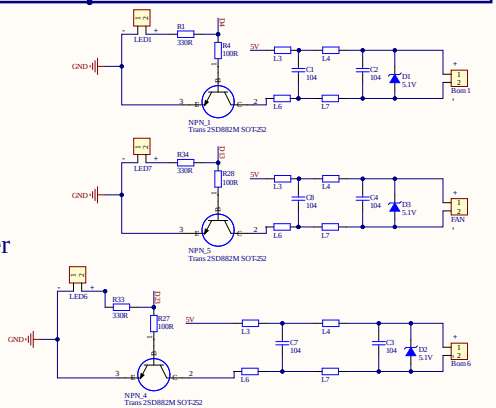


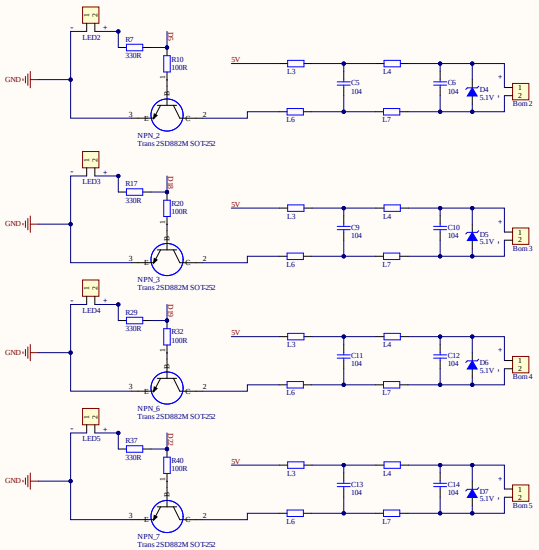
Microcontroller





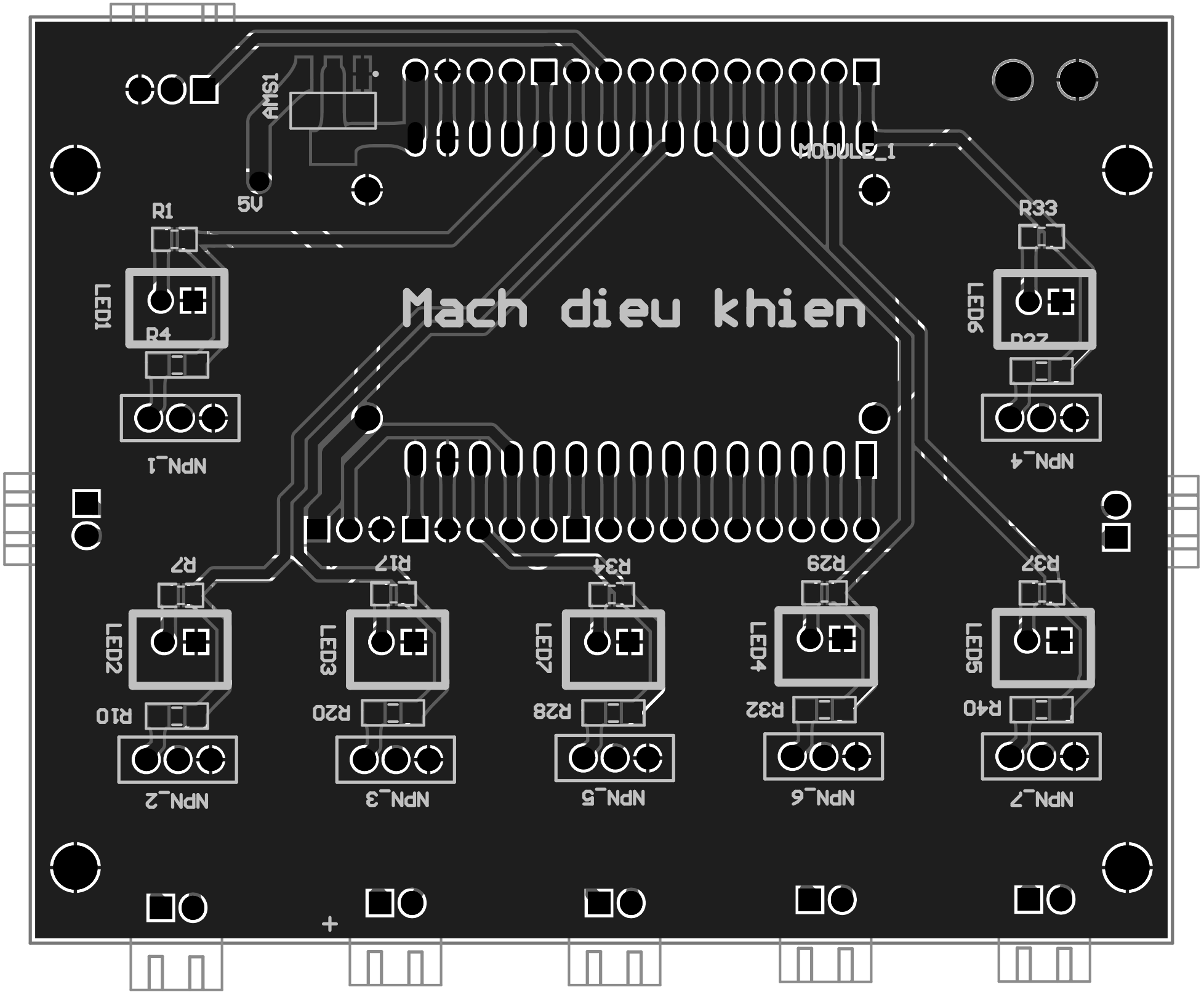
Jump + Header





Pump + Fan

Title		
Size: A	Number	Revision
Date: 11-05-2026		Sheet of
File: E:\Altium_Practice\..\Mach_sche.SchDoc		Drawn By:


Mach dieu khien
AMS1
5V
MODULE_1
R1
R33
LED1
LED6
R4
R37
NPN_1
NPN_4
R7
R17
R34
R29
R37
LED2
LED3
LED7
LED4
LED5
R10
R20
R28
R32
R40
NPN_2
NPN_3
NPN_5
NPN_6
NPN_7

Comment	Description	Designator	Footprint	LibRef	Quantity
AMS1117-3.3V	Dán	AMS1	SOT223	AMS1117 SMD	1
104	Dán	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14	tu 1206	Tu 1206 104 50V 20%	14
5.1V	Dán	D1, D2, D3, D4, D5, D6, D7	DIODE ZENNER SMD SOD80	Diode Zener 12V	7
ESP32 DEVKIT		ESP1	ESP32 WROOM 32 KIT	ESP32 WROOM 32 KIT	1
JP UART	Cắm	H1	Header 3P DC 2.54	Header 3P 2.54 Duc Cong	1
LED1	Cắm	H2	XH2.54M-2	Header 2P 2.54 Duc Thang	1
Bom 1	Cắm	H3	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
Bom 6	Cắm	H4	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
FAN	Cắm	H5	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
LED2	Cắm	H6	XH2.54M-2	Header 2P 2.54 Duc Thang	1
Bom 2	Cắm	H7	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
LED3	Cắm	H8	XH2.54M-2	Header 2P 2.54 Duc Thang	1
Bom 3	Cắm	H9	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
LED4	Cắm	H10	XH2.54M-2	Header 2P 2.54 Duc Thang	1
Bom 4	Cắm	H11	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
LED6	Cắm	H12	XH2.54M-2	Header 2P 2.54 Duc Thang	1
LED7	Cắm	H13	XH2.54M-2	Header 2P 2.54 Duc Thang	1
LED5	Cắm	H14	XH2.54M-2	Header 2P 2.54 Duc Thang	1
Bom 5	Cắm	H15	Header 2P DC 2.54	Header 2P 2.54 Duc Cong	1
JP 1x10 2.54	Cắm	JP1, JP3	JP 1x10 DT	JP 1x10 2.54 Duc Thang	2
JP 1x5 2.54	Cắm	JP4, JP5	JP 1x5 DT	JP 1x5 2.54 Duc Thang	2
JP LIGHT 220V	Cắm	JP6	JP 1x3 DT	JP 1x3 2.54 Duc Thang	1
CON 2 KF301	Cắm	KF1	Con 2 KF301 5.08	CON 2 KF301	1
LM2596HVS_MODULE		MODULE_1	LM2596HVS_MODULE	LM2596HVS_MODULE	1
Trans 2SD882M SOT-252		NPN_1, NPN_2, NPN_3, NPN_4, NPN_5, NPN_6, NPN_7	Trans_2SD882_DIP	Trans 2SD882M SOT-252	7
330R	Dán	R1, R7, R17, R29, R33, R34, R37	Tro SMD 0805	Tro 0805 5% 330R	7
L3	Dán	R2, R8, R18, R21, R23, R30, R38	tro smd 1206	Tro 1206 1% 200R	7
L4	Dán	R3, R9, R13, R15, R19, R31, R39	tro smd 1206	Tro 1206 1% 200R	7
100R	Dán	R4, R10, R20, R27, R28, R32, R40	tro smd 1206	Tro 1206 1% 100R	7
L6	Dán	R5, R11, R22, R24, R25, R35, R41	tro smd 1206	Tro 1206 1% 200R	7
L7	Dán	R6, R12, R14, R16, R26, R36, R42	tro smd 1206	Tro 1206 1% 200R	7